Advanced Mathematical Techniques
Bookwork 11

1. **Problem 5.6.19:**

   (a) Planck's theory of quantized oscillators leads to an average energy

   $$\langle \varepsilon \rangle = \frac{\sum_{n=1}^{\infty} n\varepsilon_0 \exp\left(-\frac{n\varepsilon_0}{kT}\right)}{\sum_{n=0}^{\infty} \exp\left(-\frac{n\varepsilon_0}{kT}\right)}$$

   where ε_0 is a fixed energy. Identify the numerator and denominator as binomial expansions and show that the ratio is:

   $$\langle \varepsilon \rangle = \frac{\varepsilon_0}{\exp(\varepsilon_0/kT) - 1}$$

   $$x = e^{-\varepsilon_0/kT}, \quad 0 < x < 1$$

   $$\langle \varepsilon \rangle = \frac{\sum_{n=1}^{\infty} n\varepsilon_0\, e^{-n\varepsilon_0/kT}}{\sum_{n=0}^{\infty} x^n} = \frac{\varepsilon_0 \sum_{n=1}^{\infty} nx^n}{\frac{1}{1-x}} = \frac{\varepsilon_0 \cdot \frac{x}{(1-x)^2}}{\frac{1}{1-x}} = \frac{\varepsilon_0\, e^{-\varepsilon_0/kT}}{1 - e^{-\varepsilon_0/kT}} = \frac{\varepsilon_0}{e^{\varepsilon_0/kT} - 1} \checkmark$$

   (b) Show that $\langle \varepsilon \rangle$ of part (a) reduces to kT, the classical result, for $kT \gg \varepsilon_0$.

   $$\langle \varepsilon \rangle = \frac{\varepsilon_0}{e^{\varepsilon_0/kT} - 1} \xrightarrow{u=\frac{\varepsilon_0}{kT}} \frac{\varepsilon_0}{e^u - 1} = \frac{\varepsilon_0}{u + \frac{u^2}{2} + \cdots} = \frac{\varepsilon_0}{u\left(1 + \frac{u}{2} + \cdots\right)} = \frac{kT}{1 + \frac{u}{2} + \cdots} \xrightarrow{u \to 0} kT \checkmark$$

2. **Problem 8.1.5:** In a Maxwellian distribution the fraction of particles with speed between v and $v + dv$ is

   $$\frac{dN}{N} = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{mv^2}{2kT}\right) v^2 dv$$

   N being the total number of particles. The average or expectation value of v^n is defined as $\langle v^n \rangle = N^{-1} \int v^n dN$. Show that

   $$\langle v^n \rangle = \left(\frac{2kT}{m}\right)^{n/2} \frac{\Gamma(\frac{n+3}{2})}{\Gamma(\frac{3}{2})}$$

   $$\langle v^n \rangle = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \int_0^{\infty} v^{n+2}\, e^{-mv^2/2kT}\, dv$$

   $$t = \frac{mv^2}{2kT}, v = \left(\frac{2kT}{m}\right)^{1/2} t^{1/2}, dv = \frac{1}{2}\left(\frac{2kT}{m}\right)^{1/2} t^{-1/2}\, dt, v^{n+2} = \left(\frac{2kT}{m}\right)^{(n+2)/2} t^{(n+2)/2}$$

   $$\int_0^{\infty} v^{n+2} e^{-mv^2/2kT}\, dv = \frac{1}{2}\left(\frac{2kT}{m}\right)^{(n+3)/2} \int_0^{\infty} t^{\frac{n+3}{2}-1} e^{-t}\, dt = \frac{1}{2}\left(\frac{2kT}{m}\right)^{(n+3)/2} \Gamma\left(\frac{n+3}{2}\right)$$

   $$\langle v^n \rangle = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \cdot \frac{1}{2}\left(\frac{2kT}{m}\right)^{(n+3)/2} \Gamma\left(\frac{n+3}{2}\right) = \frac{2}{\sqrt{\pi}}\left(\frac{2kT}{m}\right)^{n/2} \Gamma\left(\frac{n+3}{2}\right)$$

   $$\therefore\ \langle v^n \rangle = \left(\frac{2kT}{m}\right)^{n/2} \frac{\Gamma\left(\frac{n+3}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} \qquad \checkmark$$

3. **Problem 8.2.16:** The neutrino energy density (Fermi distribution) in the early history of the universe is given by

$$\rho_\nu = \frac{4\pi}{h^3}\int_0^\infty \frac{x^3}{\exp\left(\frac{x}{kT}\right)+1}dx$$

Show that

$$\rho_\nu = \frac{7\pi^5}{30h^3}(kT)^4$$

$$\begin{aligned}\rho_\nu = \frac{4\pi}{h^3}\int_0^\infty \frac{x^3}{e^{x/kT}+1}\,dx &\xrightarrow{u=x/kT} \frac{4\pi}{h^3}(kT)^4\int_0^\infty \frac{u^3}{e^u+1}\,du \\ &= \frac{4\pi}{h^3}(kT)^4(1-2^{-3})\Gamma(4)\zeta(4) = \frac{7\pi^5}{30h^3}(kT)^4 \checkmark\end{aligned}$$